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M PHARM
(SEM I) THEORY EXAMINATION 2022-23
DRUG DELIVERY SYSTEM

Time: 3 Hours**Total Marks: 75****Note:** Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.**10 x 2 = 20**

- (a) What do you mean by Telepharmacy?
- (b) Define pharmacogenetics.
- (c) Briefly explain the principle of feedback regulated drug delivery systems.
- (d) What is enzyme activated drug delivery system?
- (e) Enlist approaches to extend GI transit.
- (f) Define buccal strips.
- (g) What are Occuserts?
- (h) Write requirements of ophthalmic drug delivery system.
- (i) Give two examples of permeation enhancers.
- (j) Briefly explain uptake of antigens.

SECTION B

2. Attempt any two parts of the following:**2 x 10 = 20**

- (a) Explain the various approaches to formulate gastro retentive drug delivery systems.
- (b) Enumerate the characteristics of drugs to be formulated as Transdermal Drug Delivery System. Explain formulation and evaluation in detail
- (c) Describe the various formulation approaches for delivery of proteins and macromolecules.

SECTION C

3. Attempt any five parts of the following:**7 x 5 = 35**

- (a) Write a note on 3D printing of pharmaceuticals.
- (b) Explain various barriers to ocular drug delivery in detail.
- (c) Discuss mechanism of Osmotic activated drug delivery system along with examples.
- (d) Explain single shot vaccine in detail.
- (e) Elaborate on physicochemical & biological approaches for SR/CR formulations.
- (f) Explain various barriers which affect dermal permeation of drugs.
- (g) Write a note on mucosal and transdermal delivery of vaccines.